Lab : Build VPC and Launch a Web Server

Lab overview

In this lab, you will use Amazon Virtual Private Cloud (VPC) to create your own VPC and add additional components to produce a customized network. You will also create a security group. You will then configure and customize an EC2 instance to run a web server and you will launch the EC2 instance to run in a subnet in the VPC.

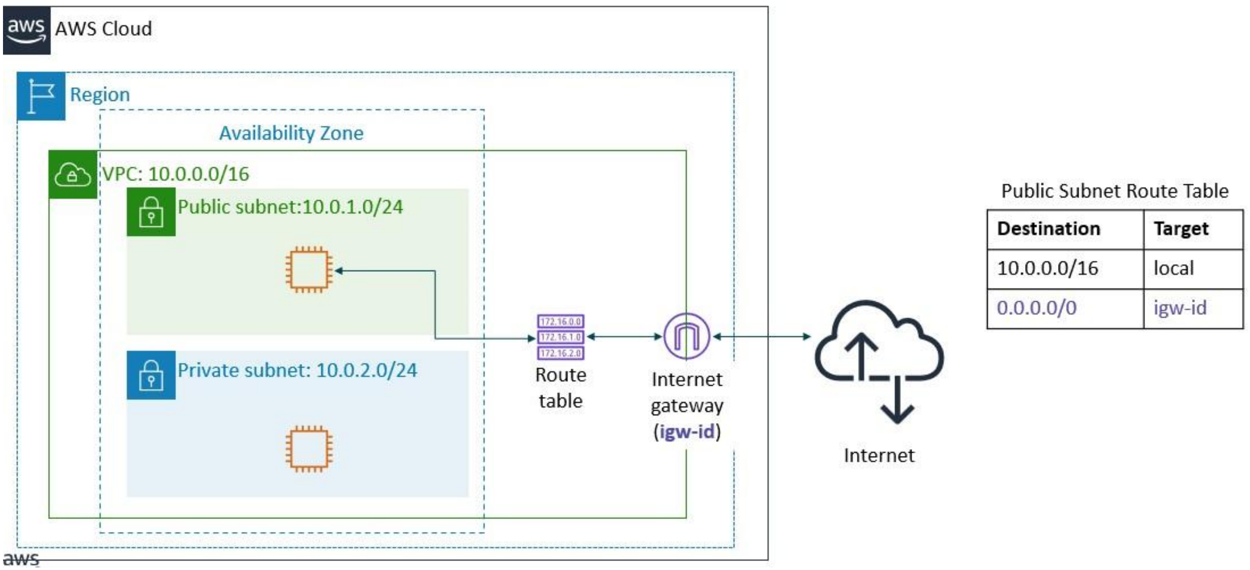
Amazon Virtual Private Cloud (Amazon VPC) enables you to launch Amazon Web Services (AWS) resources into a virtual network that you defined. This virtual network closely resembles a traditional network that you would operate in your own data center, with the benefits of using the scalable infrastructure of AWS. You can create a VPC that spans multiple Availability Zones.

After completing this lab, you should be able to do the following:

- Create a VPC.
- Create subnets.
- Launch an EC2 instance into a VPC.

Scenario

In this lab you build the following infrastructure:



Task 1: Create Your VPC

In this task, you will use the VPC and more option in the VPC console to create multiple resources, including a VPC, an Internet Gateway, a public subnet and a private subnet in a single Availability Zone, two route tables, and a NAT Gateway.

1. In the search box to the right of **Services**, search for and choose **VPC** to open the VPC console.

2. Begin creating a VPC.

   - In the top right of the screen, verify that **N. Virginia (us-east-1)** is the region.
   - Choose the **VPC dashboard** link which is towards the top left of the console.
   - Next, choose **Create VPC**.

   Note: If you do not see a button with that name, choose the Launch VPC Wizard button instead.

3. Configure the VPC details in the VPC settings panel on the left:

   - Choose **VPC and more**.
   - Under Name tag, change the value from project to **lab-vpc**.
   - Keep the IPv4 CIDR block set to 10.0.0.0/16 For
   - Number of **Availability Zones**, choose 1.
   - For Number of **public subnets**, keep the 1 setting.
   - For Number of **private subnets**, keep the 1 setting.
   - Expand the Customize subnets CIDR blocks section

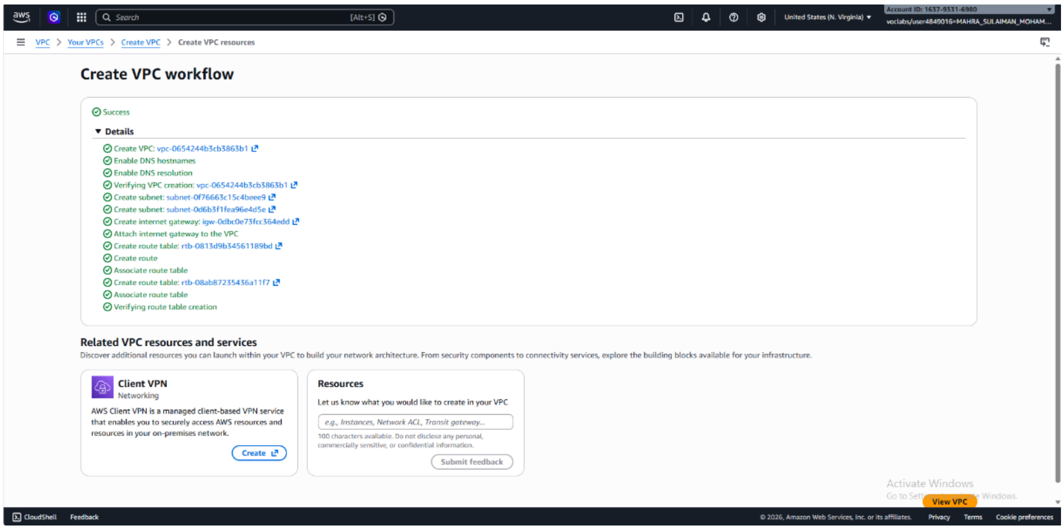
     • Change Public subnet CIDR block in us-east-1a to 10.0.1.0/24
     • Change Private subnet CIDR block in us-east-1a to 10.0.2.0/24 Set **NAT gateways**
   - to None.
   - Set **VPC endpoints** to None.
   - Keep both **DNS hostnames** and **DNS resolution** enabled.

4. In the Preview panel on the right, confirm the settings you have configured.

5. At the bottom of the screen, choose **Create VPC**

The VPC resources are created. The NAT Gateway will take a few minutes to activate.

Please wait until all the resources are created before proceeding to the next step.



6. Once it is complete, choose View VPC

The wizard has provisioned a VPC with a public subnet and a private subnet in one Availability Zone with route tables for each subnet. It also created an Internet Gateway.

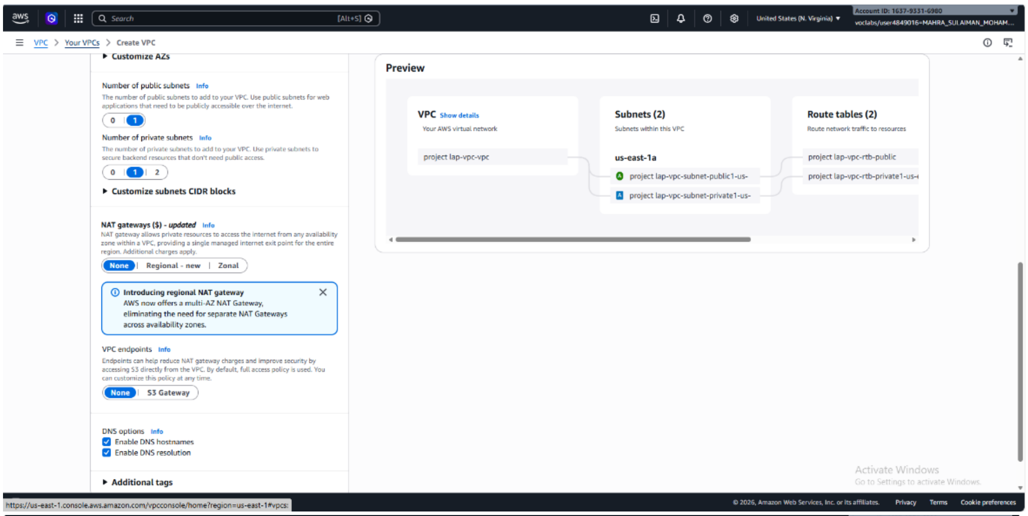
To view the settings of these resources, browse through the VPC console links that display the resource details. For example, choose Subnets to view the subnet details and choose Route tables to view the route table details.

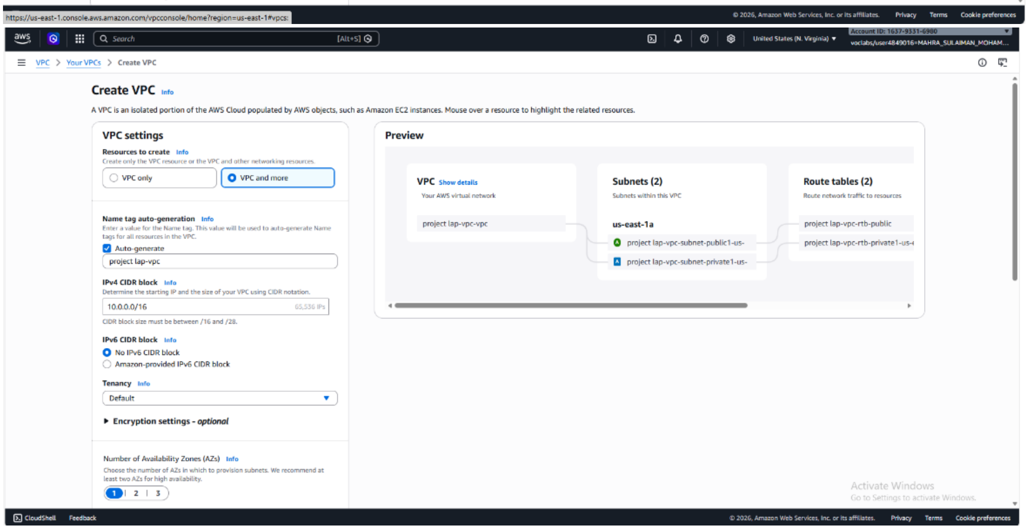
An Internet gateway is a VPC resource that allows communication between EC2 instances in your VPC and the Internet.

The lab-subnet-public1-us-east-1a public subnet has a CIDR of 10.0.1.0/24, which means that it contains all IP addresses starting with 10.0.1.x. The fact the route table associated with this public subnet routes 0.0.0.0/0 network traffic to the internet gateway is what makes it a public subnet.

A NAT Gateway, is a VPC resource used to provide internet connectivity to any EC2 instances running in private subnets in the VPC without those EC2 instances needing to have a direct connection to the internet gateway.

The lab-subnet-private1-us-east-1a private subnet has a CIDR of 10.0.2.0/24, which means that it contains all IP addresses starting with 10.0.2.x.





Task 2: Launch a Web Server Instance

In this task, you will launch an Amazon EC2 instance into the new VPC. You will configure the instance to act as a web server.

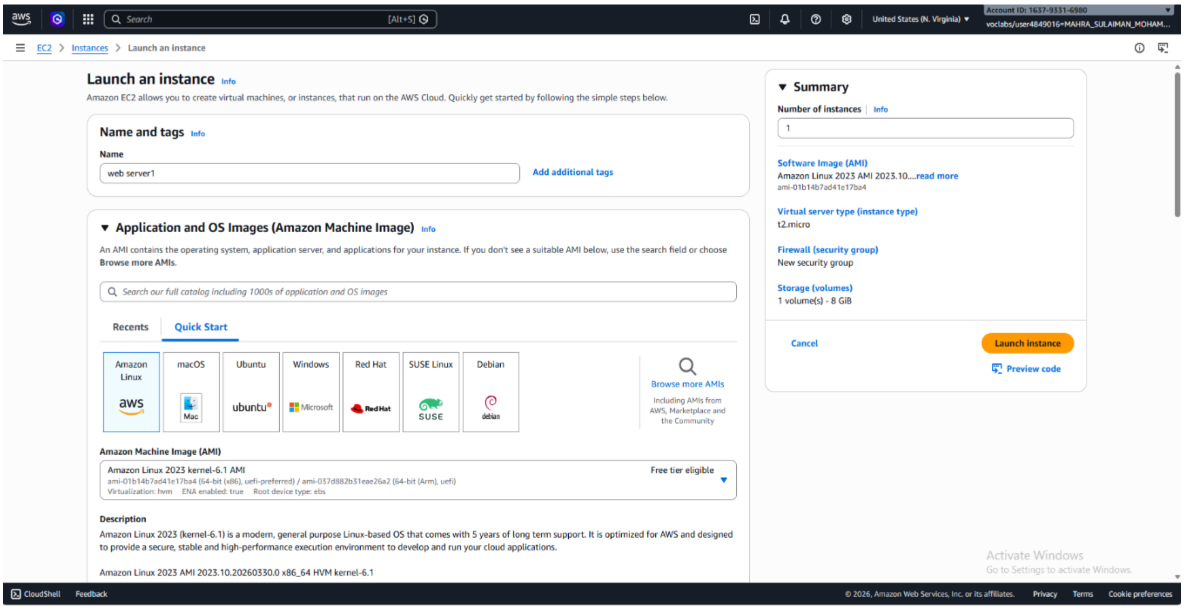
1. In the search box to the right of Services, search for and choose EC2 to open the EC2 console.

2. From the Launch instance menu choose Launch instance.

3. Name the instance:

   Give it the name Web Server 1

   When you name your instance, AWS creates a tag and associates it with the instance. A tag is a key value pair. The key for this pair is *Name*, and the value is the name you enter for your EC2 instance.



4. Choose an AMI from which to create the instance:

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   ○ In the list of available Quick Start AMIs, keep the default Amazon Linux selected.
   Select **Amazon Linux 2 AMI.**

   The type of Amazon Machine Image (AMI) you choose determines the Operating System that will run on the EC2 instance that you launch.

5. Choose an Instance type:

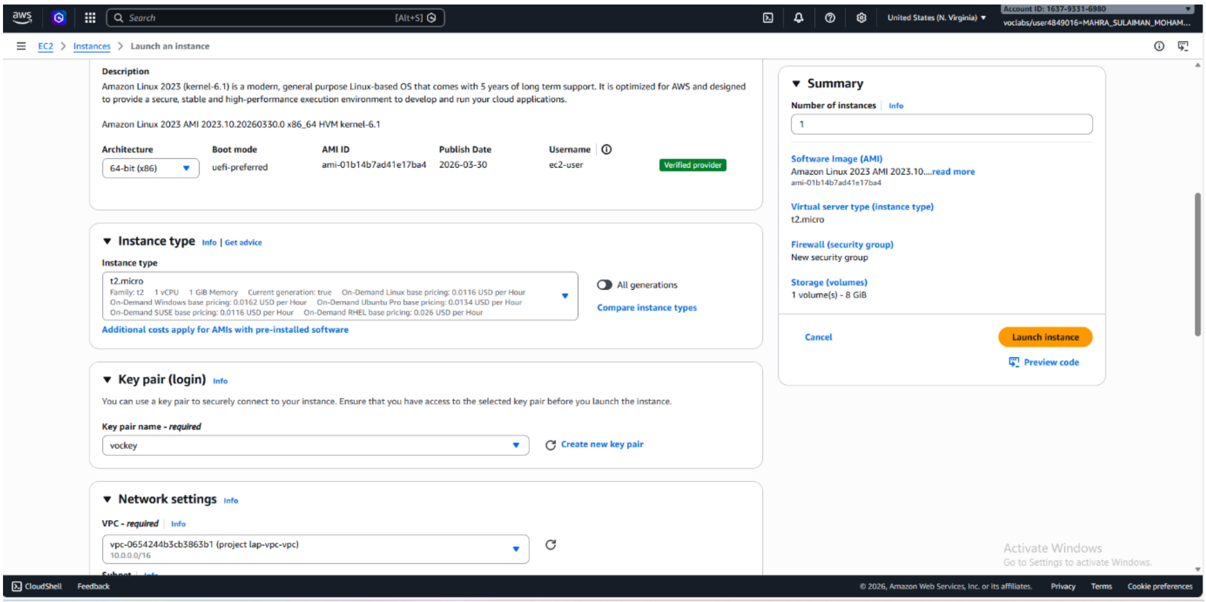
   In the Instance type panel, keep the default t2.micro selected.

   The Instance Type defines the hardware resources assigned to the instance.

6. Select the key pair to associate with the instance:

   From the Key pair name menu, select vockey.

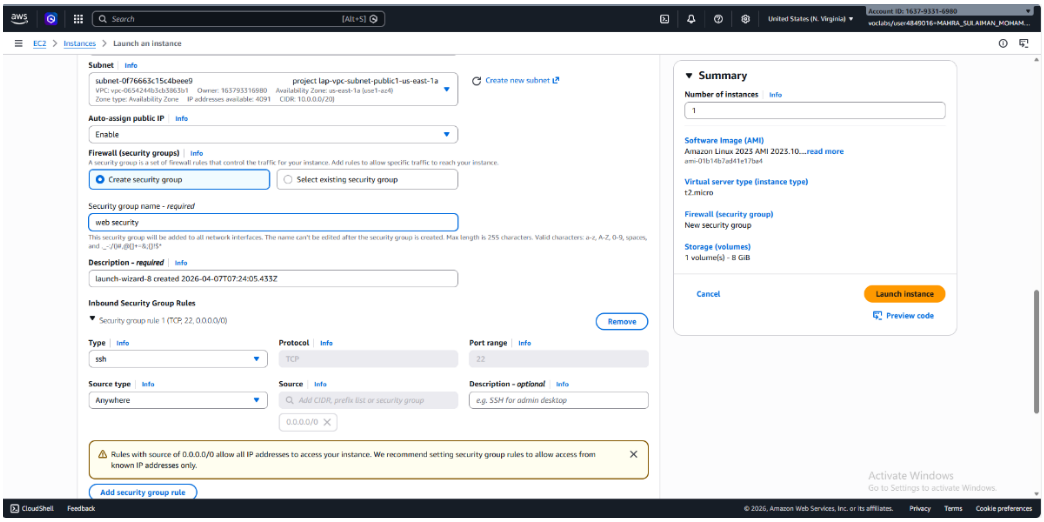
   The vockey key pair you selected will allow you to connect to this instance via SSH after it has launched. Although you will not need to do that in this lab, it is still required to identify an existing key pair, or create a new one, or choose to proceed without a key pair, when you launch an instance.



7. Configure the Network settings:

Next to Network settings, choose Edit, then configure:

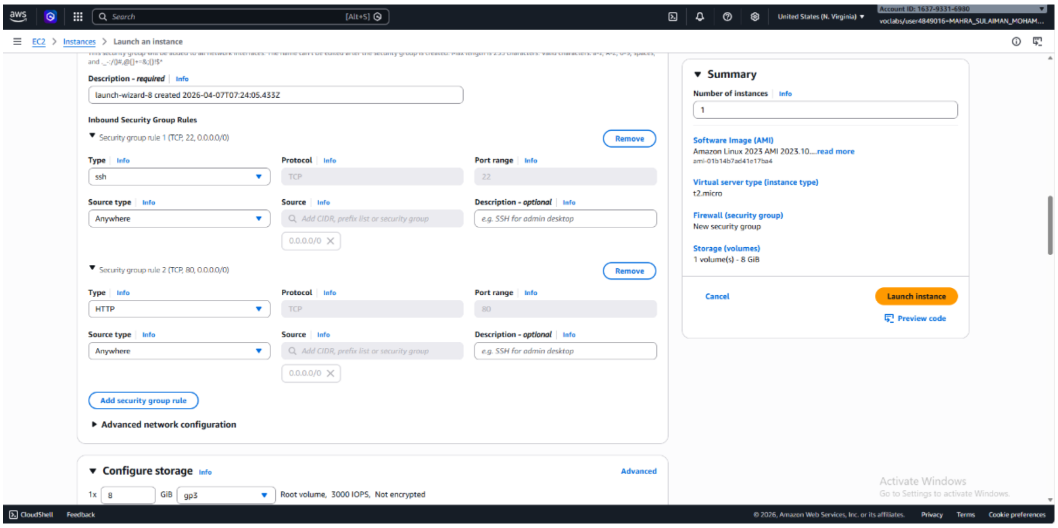
- Network: **lab-vpc**
- Subnet: lab-subnet-**public** (not Private!)
- **Auto-assign public IP: Enable**



8. Next, you will configure the instance to use the Web Security Group:

- Keep the default selection **Create a new security group**.
- **Security group name:** Clear the text and enter `Web Security`
- **Description:** Clear the text and enter `Security group for my web server`

- Keep the default selection SSH inbound rule.
- Allow HTTP and HTTPS traffic from the internet In the Inbound rules pane, choose Add rule Configure the following settings:
- Type: HTTP
- Source: Anywhere-IPv4



9. In the Configure storage section, keep the default settings.

Note: The default settings specify that the root volume of the instance, which will host the Amazon Linux guest operating system that you specified earlier, will run on a general-purpose SSD (gp3) hard drive that is 8 GiB in size.

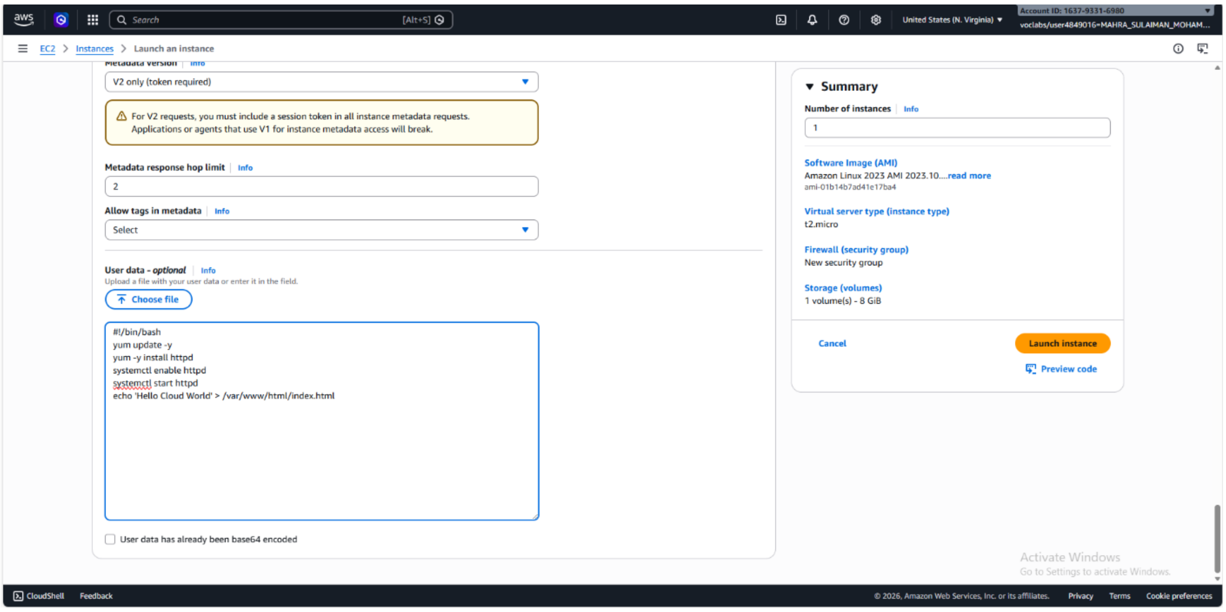
10. Configure a script to run on the instance when it launches:

    ◦

    ◦ Expand the Advanced details panel.
    Scroll to the bottom of the page and then copy and paste the code shown below into the User data

   box: xxxxxxxxxx

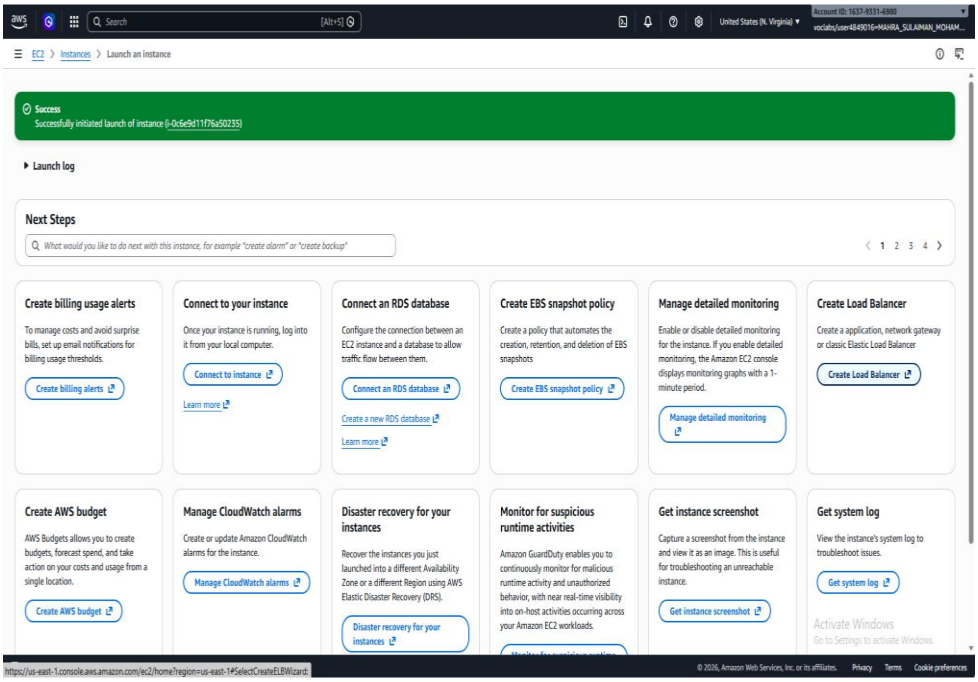
```
#!/bin/bash  yum update -y  yum -y install httpd

systemctl enable httpd  systemctl start httpd  echo

'Hello Cloud World' > /var/www/html/index.html
```

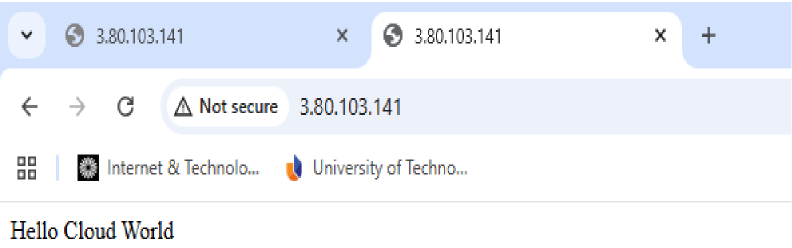


This script will run with root user permissions on the guest OS of the instance. It will run automatically when the instance launches for the first time. The script installs a web server.

11. At the bottom of the Summary panel on the right side of the screen choose Launch instance

You will see a Success message.

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